

SHREEKUMAR B

AI Engineer | Computer Vision & OCR Pipelines | Verifiable ML Systems

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Summary

B.Tech student in AI & Data Science building production-minded vision and AI systems: multi-engine OCR on noisy handwritten scans, vision-extraction pipelines, and deterministic verification layers around LLMs. Works from first principles and validates every pipeline with automated tests rather than demo runs. Global Rank #51 in HackerRank Orchestrate (Sep 2026); Team Lead of The Ragnarok.

Education

Sri Shakthi Institute of Engineering and Technology

Coimbatore, India

B.Tech in Artificial Intelligence & Data Science CGPA: 8.8/10

Expected 2029

Technical Skills

Languages: Python, SQL, JavaScript

Computer Vision & OCR: YOLOv8, ByteTrack, OpenCV, PaddleOCR, EasyOCR, Tesseract, Gemini Vision, Scikit-learn, NumPy, Pandas

AI & LLMs: LangChain, LangGraph, RAG, MCP, AI Agents, OpenAI / Gemini / Claude APIs, Ollama, ChromaDB, Tree-sitter (AST verification)

Backend: FastAPI, Fastify, Node.js, Express.js, SQLAlchemy, Drizzle ORM, BullMQ, Celery, Valkey/Redis, Zod, GitHub Webhooks, OpenTelemetry, Pytest, Vitest

Frontend: React, Next.js, Tailwind CSS, TanStack Query, shadcn/ui, Framer Motion

Databases: PostgreSQL (RLS, pgvector), SQLite, MongoDB, ChromaDB, Valkey/Redis

3D & Graphics: Blender, Unreal Engine

DevOps & Tools: Git, GitHub Actions (CI), Docker, pnpm/Turbo, uv, Linux, npm, Vercel, Google Colab, Postman

Projects

GradeMIND | *Handwritten Answer-Sheet Grader* — *FastAPI, Next.js 14, PaddleOCR, EasyOCR, Tesseract* [GitHub](#)

- Fused three OCR engines via **confidence voting** with Gemini Vision for noisy handwritten scans; built question segmentation and annotated-PDF output tracing every mark to an evidence span.
- Enforced “the model reads, arithmetic decides”: a deterministic scorer with reproducibility tests, atomic job-state persistence with resume/cache reuse, and human-override preservation.
- Led a production-readiness audit that exposed fabricated agent results and a score-desync bug; authored a six-phase remediation plan gating real student data behind human review.

PRYSM | *Continuous AI Compliance OS* — *Fastify, Next.js, FastAPI, Postgres/RLS, BullMQ* [GitHub](#)

- Architecting a multi-tenant compliance platform that turns AI regulations into versioned controls, enforces them inline via an **LLM gateway**, and keeps a **hash-chained, tamper-evident evidence trail** verifiable offline.
- Set the design rules: LLM findings never close a control (human confirmation required), every obligation must cite exact source text (checked by code), and Postgres RLS plus app-layer tenant isolation; recorded in **11 ADRs**.
- Built the M0 monorepo (pnpm/Turbo + uv; 5 services, Drizzle/SQL migrations, Valkey, OpenTelemetry) with CI gates for wording rules and TODO tracking.

AHAL AI | *Engineering Risk Intelligence Platform* — *multi-agent, Tree-sitter, GitHub webhooks* [GitHub](#)

- Designed a platform for **PR blast-radius prediction**, production root-cause diagnosis, and staged autonomous remediation, with multi-agent repository ingestion and GitHub webhook automation.
- Gated every LLM-proposed change behind **deterministic Tree-sitter/AST verification** (no LLM-judges-LLM); authored the technical whitepaper and benchmarked the design against a competing multi-agent code-review system.

DriftCheck | *Open-Source CLI for Post-Deploy Verification* — *TypeScript, Node.js, npm, GitHub Actions* [GitHub](#)

- Published to npm ([@shreekumar007/driftcheck](#)): verifies that a **live deployment actually works**, catching failures that a green build hides.
- Shipped with CI, a deliberately broken Vercel fixture as a negative test, and **91 passing tests**.

Minchal | *Photo-First Electricity Bill Attribution* — *Team Lead, HackXelerate 26' (National Finalist)*

- Designed a Gemini-extract → deterministic-calculate → Gemini-explain pipeline that estimates which household appliance drives a utility bill from photographed bills and appliance nameplates alone.

Achievements

- HackerRank Orchestrate** — Global Rank #51 (Sep 2026, score 70.7/100) | Global Rank #209 (Aug 2026)
- Winner** — Crowdera UI/UX Hackathon
- Finalist** — HackXelerate 26' (National-Level) | **Semifinalist** — International Hackathon Farway, Delhi
- Team Lead** — The Ragnarok; organized a college-level hackathon

Certifications

- Building Adaptive Agents | Vector Databases: From Embeddings to Applications | Prompt Engineering for Everyone — DeepLearning.AI
- AI Fluency — Anthropic | Introduction to OpenCV | Introduction to Machine Learning | Fundamentals of Machine Learning